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Original Contribution

Preoperative SGLT2i therapy and acute kidney injury in patients undergoing emergency and urgent coronary artery bypass grafting - A causal inference framework

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HIGHLIGHTS

- Preoperative SGLT2i linked to reduced CSA-AKI in urgent/emergency CABG.
- Causal inference framework applied to emulate randomized trial conditions.
- Protective effect most evident in HFrEF and high EuroSCORE II patients.
- No increase in postoperative DKA or adverse safety signals observed.
- Supports continuation of SGLT2i in high-risk cardiac surgery patients.

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ABSTRACT

Background: Acute kidney injury (AKI) is a common complication after coronary artery bypass grafting (CABG), associated with adverse short- and long-term outcomes. Sodium-glucose cotransporter-2 inhibitors (SGLT2i) have been shown to reduce occurrence of AKI in several populations, yet their perioperative effects in patients undergoing CABG are unknown.

Methods: We conducted a retrospective study at the Department of Cardiac Surgery, Medical University of Graz (2020–2024) to evaluate the impact of preoperative SGLT2i use on cardiac surgery–associated AKI in adults undergoing urgent or emergent isolated coronary artery bypass grafting in patients with an indication for SGLT2i therapy (type 2 diabetes mellitus, heart failure with reduced ejection fraction, or chronic kidney disease). Patients with preoperative dialysis, sepsis, reoperation, mechanical circulatory support or missing laboratory data were excluded. Exposure was defined as SGLT2i use within two weeks before surgery, and the primary outcome was cardiac surgery-associated AKI (CSA-AKI) according to KDIGO criteria. Secondary outcomes included kidney replacement therapy, ICU length of stay, 30-day mortality and postoperative diabetic ketoacidosis. Causal effects were estimated using entropy balancing. Results were reported as weighted risk differences, risk ratios, and adjusted mean differences, with time-to-event outcomes analyzed via weighted Cox models and Kaplan–Meier estimates.

Results: Among 484 patients, 135 were on SGLT2i. CSA-AKI occurred in 23.0 % of SGLT2i users vs. 28.1 % of non-users (risk ratio of 0.63 [95 % CI 0.44–0.91; $p = 0.014$]). The association was pronounced in patients with heart failure with reduced ejection fraction and those with high EuroSCORE II. No differences were observed in other secondary endpoints and no cases of postoperative diabetic ketoacidosis occurred.

Conclusion: Preoperative SGLT2i use was associated with a significantly lower risk of CSA-AKI in patients undergoing urgent or emergent CABG. These findings need to be confirmed in prospective multicenter trials but underline the favorable safety profile of this medication.

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1. Introduction

Acute kidney injury (AKI) is a frequent complication after cardiac surgery, with reported incidence rates ranging from 20 % to 30 % [1,2]. Cardiac surgery-associated AKI (CSA-AKI) is linked to increased postoperative morbidity and mortality and prolonged hospital stays [1,3]. It also carries long-term risks, including progression to chronic kidney disease (CKD) [4] and reduced life expectancy compared to patients without CSA-AKI [5]. Therefore, efforts should focus on prevention.

Sodium-glucose cotransporter-2 inhibitors (SGLT2i) are well-established therapies for type 2 diabetes mellitus (T2DM), heart failure, and CKD, and have consistently been associated with reduced rates of AKI in these populations [6–8]. However, their safety in the perioperative context, particularly prior to cardiac surgery, remains uncertain.

Current guidelines commonly recommend withholding SGLT2i a few days prior to surgery [9], yet emerging data challenge this approach, as a large cohort study found no increased risk of postoperative diabetic ketoacidosis (DKA) in patients undergoing emergency surgery, reassuring the safety profile of this medication [10]. Evidence regarding cardiac surgery, especially with respect to CSA-AKI, is limited in patients on SGLT2i therapy.

Given the lack of prospective trials in patients undergoing cardiac surgery while on SGLT2i medication, clinicians currently rely on limited retrospective data. The reliability of such studies can be improved by applying a causal inference framework, which enables observational data to be analyzed in a manner that more closely approximates the structure of a randomized trial and reduces common sources of bias [11,12].

Hence, this study aimed to evaluate the association between preoperative SGLT2i use and the risk of CSA-AKI using a causal inference framework.

2. Methods

2.1. Study design

This study was a retrospective study conducted at the Department of Cardiac Surgery of the Medical University of Graz between 2020 and 2024. Ethical approval was sought and granted by the Ethics Committee of the Medical University of Graz (IRB00001153) prior to data retrieval and study conduction. The requirement for informed consent was waived due to the retrospective nature of the analysis. Data were obtained from the electronic medical records, including preoperative anesthesia assessment documents, intraoperative anesthesia protocols, and intensive care unit (ICU)-protocols.

The study followed the STrengthening the Reporting of OBservational Studies in Epidemiology (STROBE) reporting guideline [13].

2.2. Eligibility criteria and exposure

We screened all patients who underwent isolated coronary artery bypass grafting (CABG) at the Department of Cardiac Surgery, Medical University of Graz, between 2020 and 2024. Only patients with a documented diagnosis of T2DM (excluding type 1 diabetes), heart failure with reduced ejection fraction (HFrEF), or CKD – current guideline-approved indications for SGLT2i therapy - undergoing urgent (<48 h) or emergent (within 12 h) surgery were included, under the assumption that SGLT2i therapy would not have been discontinued preoperatively in these settings. Patients were excluded if they were younger than 18 years of age, had missing key laboratory data required for AKI definition, received mechanical circulatory support therapy after surgery, had preoperative KRT, required reoperation within 72 h postoperatively, or had sepsis or septic shock prior to surgery.

Exposure was defined as the use of SGLT2i within two weeks prior to surgery, as determined through the electronic medical prescription database and preoperative anesthesia records.

This window was selected to ensure reliable classification of preoperative users in the real-world urgent and emergent setting, where exact stop dates are rarely documented. Because time-critical revascularization rarely allows for guideline-recommended discontinuation, restricting the cohort to urgent (<48 h) and emergent (≤12 h) CABG maximizes the likelihood that individuals with recorded recent use had taken their last dose shortly before surgery. Previous pharmacodynamic data have shown that measurable effects, such as glucosuria, typically decline over 12–18 h after administration, supporting the assumption that only patients undergoing surgery soon after presentation are likely to retain meaningful perioperative drug activity [14]. By combining a documentation window that minimizes misclassification with a surgical timing window that preserves recency of intake, this operational definition was designed to most accurately capture patients with genuine physiological exposure to SGLT2 inhibitors at the time of surgery.

In Austria, available SGLT2i include dapagliflozin (Forxiga®) and empagliflozin (Jardiance®).

2.3. Outcome definitions

The primary outcome was the occurrence of CSA-AKI, assessed using daily in-hospital serum creatinine measurements and defined according to the Kidney Disease: Improving Global Outcomes (KDIGO) AKI criteria [15]. CSA-AKI included all KDIGO stages within 7 days: stage 1 was defined as an increase in serum creatinine of ≥0.3 mg/dL within 48 h or a ≥ 50 % increase from baseline within seven days; stage 2 and 3 were defined as increases in serum creatinine of >200 % and > 300 % from baseline, respectively. Baseline creatinine was defined as the value obtained from the preoperative laboratory evaluation. Secondary outcomes included initiation of kidney replacement therapy (KRT), ICU-length of stay and 30-day mortality. The occurrence of DKA was evaluated as a safety outcome. Normoglycemic DKA was defined using results obtained from postoperative days 1 to 7. The definition required the presence of metabolic acidosis (bicarbonate <18 mEq/L or base excess <−5 mmol/L with a high anion gap (>12 mmol/L) and glucose levels <200 mg/dL, in the absence of elevated lactate (<3 mmol/L) [8].

2.4. Statistical analysis

All analyses were conducted using R version 4.4.0 (R Foundation for Statistical Computing, Vienna, Austria) and data preprocessing was performed in Python (version 3.12).

Baseline characteristics are presented as absolute numbers with corresponding percentages for categorical variables and as medians with 25th to 75th percentiles for continuous variables. Categorical variables were compared using the chi-squared test or Fisher's exact test, as appropriate, while continuous variables were compared using the Mann–Whitney *U* test.

To estimate the causal effect of preoperative SGLT2i use on CSA-AKI, we applied entropy balancing (EBAL) targeting the average treatment effect (ATE), implemented with the *WeightIt* [16] package in R. EBAL is a covariate balancing method that directly estimates subject-specific weights such that the weighted distributions of prespecified baseline covariates are identical between treatment groups, thereby emulating the conditions of a randomized trial.

Baseline covariates were selected a priori based on clinical plausibility and prior literature: year of operation, EuroSCORE II, sex, height, weight, age, smoking status, history of hypertension, history of chronic lung disease, prior vascular stroke, dyslipidemia, preoperative creatinine, history of T2DM, history of HFrEF and presence of preoperative anemia (hemoglobin <12 g/dL in women and < 13 g/dL in men).

Diagnostic checks focused on weight stability, since EBAL enforces exact mean balance by construction. We evaluated the distribution of weights, the effective sample size (ESS), and variability measures (coefficient of variation, entropy) (see Supplementary File e1).

For the primary outcome of CSA-AKI and the secondary outcome of

KRT, weighted logistic regression models were fitted. Marginal treatment effects were calculated as the absolute difference in predicted probabilities between treated and untreated patients (risk difference) and the ratio of predicted probabilities (risk ratio). All effect estimates were accompanied by robust standard errors and 95 % confidence intervals computed directly from the weighted model using the *marginaleffects* package in R [17].

Subgroup analyses were performed in the following prespecified patient groups: CKD, HFrEF, T2DM, age ≥ 65 years, and high EuroSCORE II (defined as greater than or equal to the median value for the study population).

For 30-day all-cause mortality, a marginal structural Cox proportional hazards model was fitted using the entropy balanced weights [18]. ICU length of stay was analyzed as a continuous outcome using a weighted linear model, yielding adjusted mean differences between treatment groups with robust standard errors.

Time-to-event data were additionally visualized using weighted Kaplan–Meier estimates.

2.5. Sensitivity analysis

To evaluate whether perioperative risk factors for CSA-AKI could account for the association between preoperative SGLT2i use and CSA-AKI, we performed a sensitivity analysis in which the original entropy balancing weights were retained and the weighted logistic regression model was additionally adjusted for perioperative risk factors. These included vasoactive support (norepinephrine >0.2 μg/kg/min or any inotropic agent within 24 h of surgery), receipt of any blood product within 24 h of surgery, hypotension (≥10 min with mean arterial pressure (MAP) <65 mmHg) within 24 h of surgery [19], preoperative catheterization within 72 h, and aortic cross-clamp time. Because these covariates occur after treatment assignment, this analysis estimates a controlled direct effect conditional on intra- and early postoperative course and therefore complements the primary total effect estimand.

2.6. Perioperative management

The indication for CABG surgery was determined by the heart team through a joint decision between cardiologists and cardiac surgeons, in accordance with current clinical guidelines [20]. The surgical strategy, including the choice and number of grafts, was determined by the operating surgeon.

General anesthesia was induced with midazolam, esketamine, propofol, fentanyl, and rocuronium following peripheral venous and radial arterial cannulation, as per centre standard of care. Inhalational anesthesia with sevoflurane (2.0–2.5 %; target minimum alveolar concentration 0.8–1.0) was administered in all patients.

Cardiopulmonary bypass management followed a standardized goal-directed perfusion strategy. Pump flow was adjusted to maintain an oxygen delivery index ≥280 mL/min/m^2. Arterial perfusion pressure was maintained within a target MAP range of 50–80 mmHg, using vasoactive support when necessary with noradrenaline being the first-line vasopressor. Postoperative management, including hemodynamic monitoring, fluid balance assessment, and transfusion strategies, was performed in accordance with current guidelines and at the discretion of the attending clinicians.

3. Results

3.1. Demographics and operative characteristics of the cohort

Between 2020 and 2024, a total of 484 patients were eligible for our analysis. Within this cohort, 135 patients were prescribed with SGLT2i (Fig. 1). Among those receiving SGLT2i, 17 % (24/135) were female. Compared with patients not receiving SGLT2i, those prescribed SGLT2i had a substantially higher EuroSCORE II (4.40 [1.90–11.30] vs. 2.40 [1.50–4.80], $p < 0.001$), a higher prevalence of HFrEF (41.5 % vs. 26.1 %, $p = 0.001$), higher baseline creatinine (1.02 [0.90–1.21] vs. 0.97 [0.85–1.13], $p = 0.010$) and higher prevalence of chronic lung disease (24.4 % vs. 16.0 %, $p = 0.045$). Regarding operative characteristics, the two groups did not differ substantially, with no significant differences in the number of grafts or aortic cross-clamp time.

Baseline characteristics alongside perioperative variables of the overall cohort, stratified by SGLT2i status, are presented in Table 1.

3.2. Primary outcome

Entropy balancing resulted in a well-balanced cohort across all baseline characteristics, with only a slight reduction in effective sample size, indicating that the weighting procedure was efficient (Supplemental File e1). Postoperatively, 23.0 % (31/135) of SGLT2i users and 28.1 % (98/349) of non-users developed CSA-AKI. This corresponded to a weighted marginal risk difference of −0.117 (95 % CI −0.199 to −0.035; $p = 0.005$) and a weighted marginal risk ratio (RR) of 0.63 (95 % CI 0.44–0.91; $p = 0.014$) (Fig. 2a). The observed difference in overall AKI development between groups was largely driven by fewer KDIGO stage I events in SGLT2i users (Fig. 2b).

3.3. Secondary outcomes

At 30 days, 5 of 135 (3.7 %) SGLT2i users and 9 of 342 (2.6 %) non-users had died. This corresponded to a non-significant hazard ratio of 1.31 (95 % CI 0.47–3.63; $p = 0.535$) in weighted Cox proportional hazards regression (Fig. 3).

Postoperative KRT was required in 2 of 135 SGLT2i users (1.5 %) compared to 6 of 349 non-users (1.7 %) ($p = 0.793$).

Regarding ICU length of stay, no significant difference was observed. Median ICU stay was 3 days (IQR 2–6) in SGLT2i users and 3 days (IQR 2–5) in non-users. This corresponded to an adjusted mean difference of −0.142 days (95 % CI −1.981 to 1.703; $p = 0.881$) in weighted analysis.

3.4. Subgroup analysis

SGLT2i use was most common in patients with HFrEF (38.1 %, 56/147) and those with a high EuroSCORE II (35.5 %, 86/242), followed by patients with CKD (31.2 %, 39/125), T2DM (30.1 %, 53/176), and age ≥ 65 years (30.0 %, 80/267).

The risk of CSA-AKI was significantly lower in patients with high EuroSCORE II (RR = 0.57, 95 % CI 0.34–0.93, $p = 0.026$) and in those with HFrEF (RR = 0.50, 95 % CI 0.25–0.98, $p = 0.043$), if patients were on SGLT2i. Although reductions in risk were also observed in patients aged ≥65 years (RR = 0.65, 95 % CI 0.39–1.07, $p = 0.087$) and those with CKD (RR = 0.47, 95 % CI 0.20–1.07, $p = 0.073$), these did not reach statistical significance. No significant association was found for T2DM (RR = 0.81, 95 % CI 0.48–1.39, $p = 0.445$) (Fig. 4).

3.5. Sensitivity analysis

To assess whether the observed association between preoperative SGLT2i use and CSA-AKI could be explained by perioperative risk factors for CSA-AKI, the weighted logistic regression model with additional adjustment for vasoactive support, transfusion of any blood product within 24 h, hypotension (≥10 min with MAP <65 mmHg) within 24 h of surgery, preoperative catheterization within 72 h, and aortic cross-clamp time was refitted. After incorporating these variables, the association remained directionally consistent with the primary result. Preoperative SGLT2i use was associated with a risk difference of −0.106 (95 % CI −0.189 to −0.022; $p = 0.013$) and a RR of 0.66 (95 % CI 0.46–0.95; $p = 0.027$). These findings indicate that the reduced risk of CSA-AKI associated with SGLT2i use was not explained by differences in postoperative vasoactive requirements, transfusion exposure, or hypotensive burden.

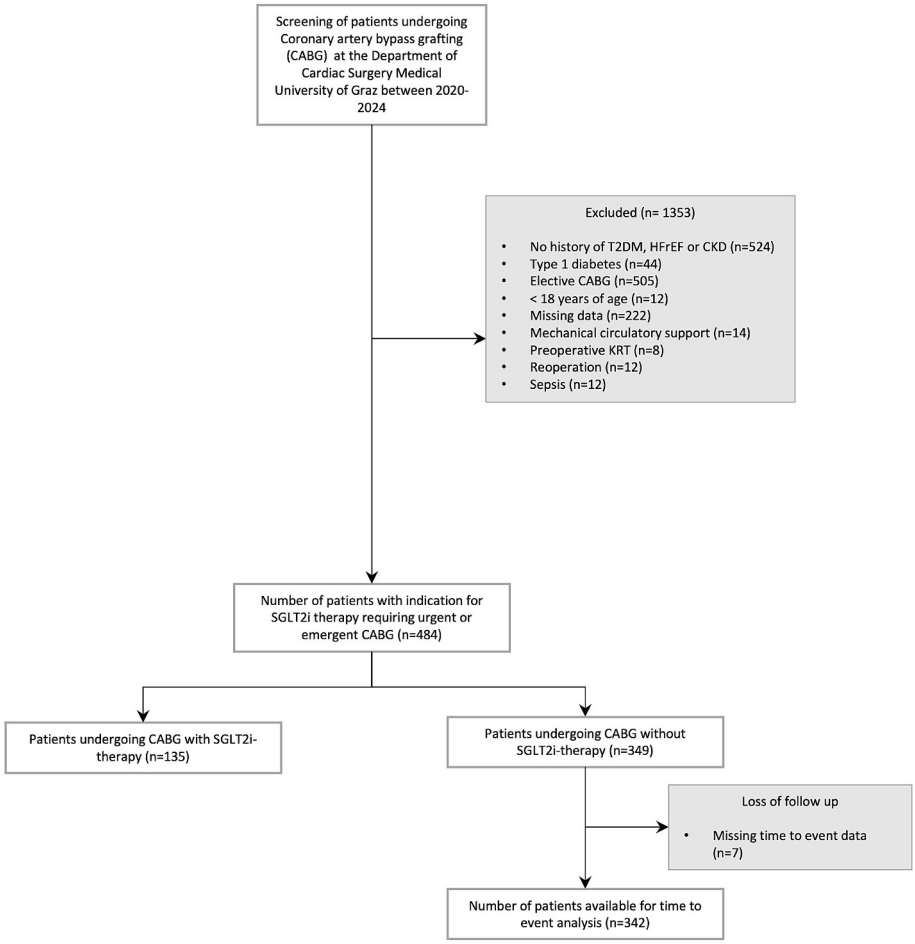


Fig. 1. Study Flow Chart.
Abbreviations: CABG coronary artery bypass graft; CKD – Chronic Kidney Disease; HFrEF – Heart failure with reduced ejection fraction; T2DM – type 2 diabetes mellitus; SGLT2i sodium–glucose cotransporter 2 (SGLT2) inhibitors.

4. Discussion

In this causal inference analysis of patients undergoing urgent or emergency CABG, preoperative SGLT2i use was associated with a significantly lower risk of postoperative CSA-AKI, while KRT, 30-day mortality and ICU length of stay did not differ between groups.

So far, high quality evidence on SGLT2i use in the perioperative setting is scarce and the key limitation in comparing existent studies is the variable timing of drug initiation and, more importantly, discontinuation. Some studies withheld SGLT2i before scheduled surgery, whereas others began treatment during or after hospitalization. This heterogeneity may lead to varying drug exposure during surgery, making it difficult to determine whether any observed kidney protection or safety effects are due to the drug itself, the duration of treatment, or underlying patient characteristics. Recently, two large-scale retrospective cohort studies examined perioperative outcomes with SGLT2i. One found a modestly increased risk of DKA in both elective and emergency procedures with a reduced incidence of AKI [8]. The other only included emergency settings and did not observe any increase in the incidence of postoperative DKA [10]. Both studies did not specifically report outcomes on CSA-AKI [8,10].

Most prospective studies focused on postoperative initiation of SGLT2i in patients undergoing cardiac surgery [21]. In a recent review using a PRISM approach, three studies on perioperative SGLT2i use in patients undergoing cardiac surgery were included [22]. Of these, only one small pilot-study, comprising 55 patients, evaluated preoperative use of a SGLT2i on postoperative biomarkers and incidence of AKI. Despite no evidence of reduction of kidney stress markers, there was a large reduction of AKI (20 % vs 66.7 %) [23]. This finding is more pronounced but consistent with our study, which is further strengthened by the application within the causal inference framework. In contrast to most of the existing literature with inconsistent timing of SGLT2i therapy, the urgent - and emergency cardiac surgery setting offers a more reliable framework for recent SGLT2i use prior to surgery, minimizing timing-related bias. Although baseline creatinine was slightly higher in the SGLT2i group, this variable was explicitly balanced in the weighted analysis, making it unlikely that the observed reduction in CSA-AKI is driven by differences in preoperative kidney function.

A key mechanism of kidney damage in CSA-AKI is renal hypoperfusion resulting in decreased oxygen delivery. Low oxygen delivery has been associated with AKI in patients undergoing cardiopulmonary bypass surgery [24]. As SGLT2i decrease active sodium resorption in the proximal tubules, local oxygen and ATP consumption is reduced, which might mitigate mitochondrial production of reactive oxygen species and, thus, oxidative stress, accompanied by altered glomerular hemodynamics and reduced hydrostatic glomerular pressure [25].

Also, CSA-AKI-induced systemic inflammation and endothelial dysfunction might be mitigated by SGLT2i, by reduced inflammatory cytokine expression, inhibition of tubular senescence and apoptosis and reduced fibrosis after ischemia [25,26]. Previously, a large randomized controlled trial showed that perioperative supplementation of amino acids (AA) reduced the incidence of AKI in patients undergoing cardiac surgery [27]. In our center, however, the standard operative protocol of intra- and perioperative care did not change during the study period and

Table 1
Baseline characteristics and perioperative variables stratified by SGLT2i use.

Variable	No SGLT2i ($n = 349$)	SGLT2i ($n = 135$)	p-value
EuroSCORE II, median [IQR]	2.40 [1.50, 4.80]	4.40 [1.90, 11.30]	<0.001
Female sex, n (%)	55 (15.8)	29 (21.5)	0.175
Height (cm), median [IQR]	174.00 [169.00, 180.00]	174.00 [169.00, 179.00]	0.636
Weight (kg), median [IQR]	85.00 [75.00, 94.00]	83.00 [75.00, 93.00]	0.552
Age (years), median [IQR]	66.00 [58.00, 72.00]	67.00 [60.00, 74.00]	0.175
Current smoker, n (%)	166 (47.6)	57 (42.2)	0.339
Hypertension, n (%)	316 (90.5)	118 (87.4)	0.395
Chronic lung disease, n (%)	56 (16.0)	33 (24.4)	0.045
History of stroke/TIA, n (%)	24 (6.9)	14 (10.4)	0.274
Dyslipidemia, n (%)	292 (83.7)	109 (80.7)	0.528
HFrEF, n (%)	91 (26.1)	56 (41.5)	0.001
Diabetes mellitus, n (%)	123 (35.2)	53 (39.3)	0.473
Preoperative anemia, n (%)	55 (15.8)	13 (9.6)	0.339
Preoperative catheterization within 72 h, n (%)	44 (12.6)	20 (14.8)	0.622
CKD (eGFR <60 mL/min/1.73m^2), n (%)	86 (24.6)	39 (28.9)	0.400
Creatinine (mg/dL), median [IQR]	0.97 [0.85, 1.13]	1.02 [0.90, 1.21]	0.010
Number of anastomoses, n (%)			0.824
1	79 (22.6)	27 (20.0)	
2	165 (47.3)	62 (45.9)	
3	102 (29.2)	45 (33.3)	
>3	3 (0.9)	1 (0.7)	
Aortic clamp time (min), median [IQR]	70.00 [53.00, 83.00]	67.00 [53.00, 84.00]	0.393
Vasoactive support within 24 h of surgery, n (%)	67 (19.2)	26 (19.3)	1.000
Transfusion of any blood product within 24 h of surgery, n (%)	156 (44.7)	50 (37)	0.154
Hypotension (≥10 min with MAP <65 mmHg) within 24 h of surgery	40 (11.5)	17 (12.6)	0.850

Abbreviations: CKD – Chronic Kidney Disease; CABG – Coronary artery bypass grafting; EuroSCORE II – European System for Cardiac Operative Risk Evaluation II; HFrEF – heart failure with reduced ejection fraction; IQR – Interquartile range; TIA – Transient ischemic attack; mg/dL – milligrams per deciliter; min – minutes.

AA were not systematically administered, consequently, the impact of this protective measure on our AKI incidence is likely negligible.

There are multiple possible explanations why specifically in our study patients with higher EuroSCORE II and known congestive heart failure benefited most from prior exposure to SGLT2i in terms of occurrence of AKI. Most notably, the higher prevalence of comorbidities in patients on SGLT2i may reflect more high-risk patients, who are more susceptible to the renal, cardiovascular, and anti-inflammatory protective mechanisms of SGLT2i, potentially leading to a proportionally greater risk reduction. As the provided subgroup analyses were not mutually exclusive, some significant overlap between subgroups may have occurred. Therefore, these findings should be regarded as

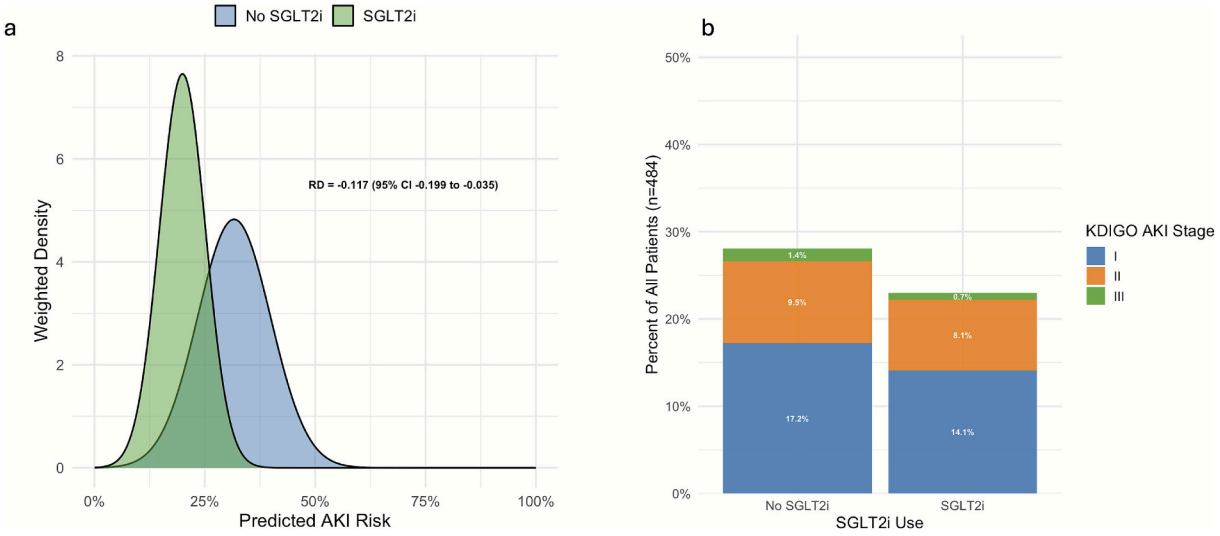


Fig. 2. a) Predicted CSA-AKI risk
Weighted density plot of predicted postoperative AKI risk in patients undergoing CABG, stratified by SGLT2 inhibitor (SGLT2i) use. The green curve represents SGLT2i users, and the blue curve represents non-users. Risk estimates were derived from the weighted logistic regression model. The mean weighted risk difference between groups was −0.117 (95 % CI −0.199 to −0.035; $p = 0.005$), indicating a lower risk of AKI in the SGLT2i group.
b) Distribution of KDIGO AKI Stages by SGLT2i Use
Stacked bar chart showing the distribution of KDIGO AKI stages I–III among patients who developed postoperative AKI after CABG, stratified by SGLT2i use. Bars display the percentage contribution of each KDIGO stage within the AKI cases of each group. Most AKI events in both groups occurred in KDIGO stage I, with fewer stage II and III events. A weighted Rao–Scott chi-square test showed no significant difference in the distribution of KDIGO AKI stages between SGLT2i users and non-users ($p = 0.82$).
Abbreviations: AKI – Acute kidney injury; CABG coronary artery bypass graft; SGLT2i – Sodium–glucose cotransporter 2 (SGLT2) inhibitors

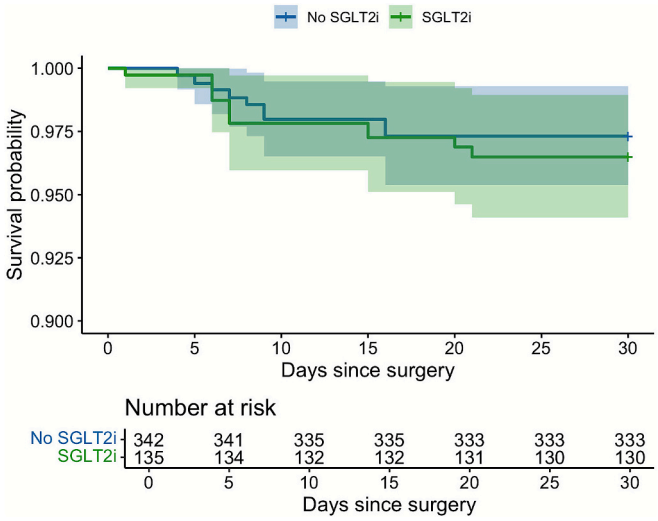


Fig. 3. Weighted Kaplan–Meier survival curves for 30-day mortality.
Weighted Kaplan–Meier survival curves for 30-day mortality following CABG, stratified by SGLT2i use. The green curve represents patients receiving SGLT2i, and the blue curve represents non-users. Shaded areas indicate 95 % confidence intervals. The number of patients at risk at each time point is shown below the x-axis. There was no statistically significant difference in 30-day survival between groups in the weighted Cox proportional hazards model. (Seven patients had missing time to event data.)
Abbreviations: CABG coronary artery bypass graft; SGLT2i – Sodium glucose transporter 2 inhibitor; (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

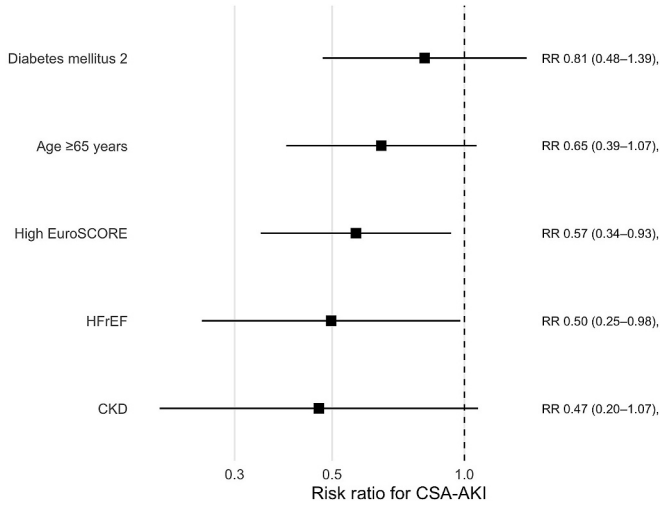


Fig. 4. Forest plot of subgroup analyses for the association between SGLT2 inhibitor use and risk of CSA-AKI.
Risk ratios with 95 % confidence intervals and corresponding *p*-values are shown for each predefined patient subgroup. Estimates are adjusted using the specified weighted logistic regression model.
Abbreviations: CKD – Chronic Kidney Disease, HFrEF – Heart failure with reduced ejection fraction.

exploratory and hypothesis-generating rather than definitive. A reason why there was no difference in initiation of KRT between groups might be the low incidence in our study cohort.

Although our results indicate that patients with higher operative risk or HFrEF may derive particular benefit from ongoing therapy, whether initiating SGLT2i preoperatively in non-emergent settings could confer similar protection remains uncertain and warrants prospective evaluation.

Currently, there is no universal agreement on the optimal time point for restarting SGLT2i therapy after major surgery [28]. In our institution, re-initiation is individualized and generally deferred until patients demonstrate hemodynamic stability and adequate oral intake. Because these decisions vary between clinicians, postoperative SGLT2i use could not be reliably or systematically captured for this analysis. Existing data suggest potential benefits when SGLT2i are initiated during postoperative recovery, and future work should further clarify how the timing of postoperative SGLT2i therapy influences renal and clinical outcomes [29].

Because of the retrospective nature, several drawbacks and limitations of this study apply. First, it is inherently subject to selection and information bias and is limited by unmeasured confounding. Second, the single-center nature may limit external validity. Third, while restricting the cohort to emergent and urgent cardiac surgery patients helps to reduce the risk of timing bias, it does not fully eliminate it. In this context, exact preoperative stopping time points for SGLT2 inhibitors were not systematically documented and therefore, exposure was defined as intake within 14 days before surgery to minimize misclassification and ensure a high likelihood of physiological drug exposure, thereby approximating a group with continued therapy as accurately as possible within the constraints of real-world data. Fourth, we can only speculate on the reasons for the positive observed effects, due to the study design. Fifth, only serum creatinine changes were used for definition of CSA-AKI, which may lead to a lower incidence of AKI [30]. Furthermore, CSA-AKI incidence varies, depending also in type of AKI definition used [31]. We did not assess long-term outcomes of kidney function which might be improved by a reduced AKI incidence as this outcome was out of the scope of this analysis. Individual SGLT2i were not analyzed separately, and dosing was not available. As CSA-AKI incidence is known to vary across hospitals and physicians, interpretation of the results is limited by the single-center nature of this study [2].

In this study, we provide evidence for continuation of SGLT2i in patients undergoing coronary artery bypass grafting as no further safety concerns occurred. Furthermore, if confirmed by prospective trials, the use of SGLT2i might be a specific nephron-protective measure which is easy to implement.

Until evidence from the currently ongoing prospective multicenter studies [32,33] is available, this study adds to the increasing body of literature that supports continuing the prescription of SGLT2i in patients with diabetes mellitus, CKD, or heart failure who undergo cardiac surgery.

CRediT authorship contribution statement

Michael Kolland: Writing – review & editing, Writing – original draft, Conceptualization. **Selina Sartori:** Writing – review & editing, Methodology, Data curation. **Christoph Klivinyi:** Writing – review & editing, Visualization, Validation, Data curation. **Michael Schörghuber:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Jakob Pannold:** Writing – review & editing, Visualization, Project administration, Methodology, Investigation, Conceptualization. **Igor Knez:** Writing – review & editing, Formal analysis, Data curation. **Alexander H. Kirsch:** Writing – review & editing, Validation, Project administration, Formal analysis, Data curation. **Nikolaus Schreiber:** Writing – original draft, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Author statement

In response to the reviewers' remarks, we have revised the manuscript extensively. In particular, we incorporated **additional sensitivity analyses directly guided by the reviewers' suggestions**, expanded and clarified methodological descriptions, refined the presentation of results, and further contextualized our findings in light of current evidence. A detailed, point-by-point response outlining all revisions

accompanies the resubmission.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclinane.2025.112103.

References

[1] Hu J, Chen R, Liu S, Yu X, Zou J, Ding X. Global incidence and outcomes of adult patients with acute kidney injury after cardiac surgery: a systematic review and Meta-analysis. J Cardiothorac Vasc Anesth 2016;30(1):82–9.
[2] Mathis MR, Mentz GB, Cao J, Balczewski EA, Janda AM, Likosky DS, et al. Hospital and clinician practice variation in cardiac surgery and postoperative acute kidney injury. JAMA Netw Open 2025;8(5):e258342.
[3] Schurle A, Koyner JL. CSA-AKI: incidence, epidemiology, clinical outcomes, and economic impact. J Clin Med 2021;10(24).
[4] Molina Andujar A, Escudero VJ, Pineiro GJ, Lucas A, Rovira I, Matute P, et al. Impact of cardiac surgery associated acute kidney injury on 1-year major adverse kidney events. Front Nephrol 2023;3:1059668.
[5] Peerapornratana S, Fiorentino M, Priyanka P, Murugan R, Kellum JA. Recovery after AKI: effects on outcomes over 15 years. J Crit Care 2023;76:154280.
[6] Zhuo M, Paik JM, Wexler DJ, Bonventre JV, Kim SC, Patorno E. SGLT2 inhibitors and the risk of acute kidney injury in older adults with type 2 diabetes. Am J Kidney Dis 2022;79(6):858–67 e1.
[7] Chung MC, Hung PH, Hsiao PJ, Wu LY, Chang CH, Hsiao KY, et al. Sodium-glucose transport protein 2 inhibitor use for type 2 diabetes and the incidence of acute kidney injury in Taiwan. JAMA Netw Open 2023;6(2):e230453.
[8] Tallarico RT, Jing B, Lu K, Chawla SA, Luo Y, Badathala A, et al. Postoperative outcomes among sodium-glucose cotransporter 2 inhibitor users. JAMA Surg 2025;160(6):681–9.
[9] Li XQ, Mirsaidi N, Bauman J, Marino J, Harris YT, Schulman-Rosenbaum R. Updates on perioperative cessation of SGLT2 inhibitors. Br J Anaesth 2025;134(5):1523–5.
[10] Dixit AA, Bateman BT, Hawn MT, Odden MC, Sun EC. Preoperative SGLT2 inhibitor use and postoperative diabetic ketoacidosis. JAMA Surg 2025;160(4):423–30.
[11] Zoccali C, Tripepi G. Clinical trial emulation in nephrology. J Nephrol 2025;38(1):11–23.
[12] Hernan MA, Wang W, Leaf DE. Target trial emulation: a framework for causal inference from observational data. JAMA 2022;328(24):2446–7.
[13] von Elm E, Altman DG, Egger M, Pocock SJ, Gotzsche PC, Vandenbroucke JP, et al. The Strengthening the reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet 2007;370(9596):1453–7.
[14] Tahara A, Takasu T, Yokono M, Imamura M, Kurosaki E. Characterization and comparison of sodium-glucose cotransporter 2 inhibitors in pharmacokinetics, pharmacodynamics, and pharmacologic effects. J Pharmacol Sci 2016;130(3):159–69.
[15] Summary of Recommendation Statements. Kidney Int Suppl 2011;3(1):5–14.
[16] Greifer N. WeightIt [cited 2025 22.09.2025]. Available from: https://ngreifer.github.io/WeightIt/reference/index.html; 2025.
[17] Arel-Bundock V, Greifer N, Heiss A. How to interpret statistical models using marginaleffects for R and Python. J Stat Softw 2024;111(9):1–32.
[18] Zhao Q, Percival D. Entropy balancing is doubly robust. J Causal Infer 2017;5(1).
[19] Ngu JMC, Jabagi H, Chung AM, Boodhwani M, Ruel M, Bourke M, et al. Defining an intraoperative hypotension threshold in association with De novo renal replacement therapy after cardiac surgery. Anesthesiology 2020;132(6):1447–57.
[20] Byrne RA, Rossello X, Coughlan JJ, Barbato E, Berry C, Chieffo A, et al. 2023 ESC guidelines for the management of acute coronary syndromes. Eur Heart J 2023;44(38):3720–826.
[21] Makortoff L, Then KL, Dutchak M, Lin M, Youngson E, Harten C. ECLIPSES: early initiation of sodium glucose cotransporter-2 inhibitors for cardiovascular protection in patients with type 2 diabetes following acute coronary syndrome and subsequent coronary artery bypass graft surgery. Pharmacotherapy 2024;44(11):841–50.
[22] Al Chalabi H, Omar AS, Gordon A, Rai B. Role of sodium-glucose transport protein 2 (SGLT2) inhibitors in preventing cardiac surgery-associated acute kidney injury: a scoping review. Cureus 2025;17(6):e86630.
[23] Snel LIP, Oosterom-Eijmael MJP, Rampanelli E, Lankadeva YR, Plummer MP, Preckel B, et al. The effects of sodium-glucose transporter 2 inhibition on cardiac surgery-associated acute kidney injury: an open-label randomized pilot study. J Clin Anesth 2025;103:111811.
[24] Ranucci M, Romitti F, Isgro G, Cotza M, Brozzi S, Boncilli A, et al. Oxygen delivery during cardiopulmonary bypass and acute renal failure after coronary operations. Ann Thorac Surg 2005;80(6):2213–20.
[25] Nakao Y., Mori M., Mori Y., Bonventre J.V. SGLT2 inhibitors and acute kidney injury. Nephrol Dial Transplant. doi:10.1093/ndt/gfaf132. Published online July 30, 2025.
[26] Zhang R, Xie Q, Lu X, Fan R, Tong N. Research advances in the anti-inflammatory effects of SGLT inhibitors in type 2 diabetes mellitus. Diabetol Metab Syndr 2024;16(1):99.
[27] Landoni G, Monaco F, Ti LK, Baiardo Redaelli M, Bradic N, Comis M, et al. A randomized trial of intravenous amino acids for kidney protection. N Engl J Med 2024;391(8):687–98.
[28] Writing Committee M, Thompson A, Fleischmann KE, Smilowitz NR, de Las Fuentes L, Mukherjee D, et al. 2024 AHA/ACC/ACS/ASNC/HRS/SCA/SCCT/SCMR/SVM guideline for perioperative cardiovascular Management for Noncardiac Surgery: a report of the American College of Cardiology/American Heart Association joint Committee on clinical practice guidelines. J Am Coll Cardiol 2024;84(19):1869–969.
[29] Al Namat R, Duceac LD, Chelaru L, Dabija MG, Gutu C, Marcu C, et al. Post-coronary artery bypass grafting outcomes of patients with/without Type-2 diabetes mellitus and chronic kidney disease treated with SGLT2 inhibitor dapagliflozin: a single-Center experience analysis. Diagnostics (Basel) 2023;14(1).
[30] Allen JC, Gardner DS, Skinner H, Harvey D, Sharman A, Devonald MAJ. Definition of hourly urine output influences reported incidence and staging of acute kidney injury. BMC Nephrol 2020;21(1):19.
[31] Brown JR, Baker RA, Shore-Lesserson L, Fox AA, Mongero LB, Lobdell KW, et al. The Society of Thoracic Surgeons/Society of Cardiovascular Anesthesiologists/American Society of Extracorporeal Technology Clinical Practice Guidelines for the prevention of adult cardiac surgery-associated acute kidney injury. Ann Thorac Surg 2023;115(1):34–42.
[32] Oosterom-Eijmael M, Monteiro de Oliveira NP, Niesten ED, Tolsma M, Snellen FT, Gerritse BM, et al. Study protocol of the multicentre, randomised, triple-blind, placebo-controlled MERCURI-2 trial: promoting effective renoprotection in cardiac surgery patients by inhibition of sodium glucose cotransporter (SGLT)-2. BMJ Open 2025;15(5):e095504.
[33] Coca A, Bustamante-Munguira E, Fidalgo V, Fernandez M, Abad C, Franco M, et al. EValuating the effect of periopeRaTIve empaGliflOzin on cardiac surgery associated acute kidney injury: rationale and design of the VERTIGO study. Clin Kidney J 2024;17(8):sfae229.